

Nutrient Sensing and Autophagy

Sérgio F genetic report
2 of October 2026 (local time)

Name	Sérgio F
Focus	longevity and immortality (84 aging mechanisms)
Data	Crude DNA (chip) + local imputation (only DR2 ≥ 0.8)
Content	12 Curated SNPs · 60 loci of GWAS · 3 polygenic scores · 5 904 pathogenic variants of ClinVar in 42 genes

What is: Sérgio F variants in the pathways that read energy and nutrients (insulin/IGF-1, mTOR, AMPK, sirtuins, FOXO) and autophagy (ATG16L1, IRGM), plus IGF-1 and 2 diabetes scores and a search for rare variants in 42 genes of these pathways.
Privacy: the analysis was done locally; the genotypes never left the computer. Public databases were only sent SNP identifiers.

Important: genetic predispositions, not a diagnosis. The conclusions come only from the genes (without analysis or exams). All practical ideas are suggestion (not applied), to decide with the doctor.

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1. Abstract: favourable and unfavourable

Favourable	Unfavourable
• FOXO3 homozygous longevity in 3 SNPs (rs2802292 GG, rs2764264 CC, rs13217795 CC): the most replicated genetic sign; FOXO3 activates autophagy, antioxidants and repair. • Genetically lower IGF-1: 52 loci score at the 7th percentile (genomic at the 34th percentile). • 14 IGF-1 loci with lower IGF-1 alleles (e.g. IGFBP3, PAPPA2, HNF1A). • ClinVar: no plausible pathogenic variant in 42 genes; the only hit (TSC2 in homozygous) is artefact.	• Heterozygotias in the medium in TCF7L2 rs7903146, FTO rs9939609, IRS1 rs2943641, MTOR rs2295080 and ATG16L1 Thr300Ala (1 allele of risk each; do not move away from the average). • SIRT1 rs7895833 AA and PPARG Pro12Pro: without the least frequent protective alleles (typical). • 5 IGF-1 loci with higher IGF-1 alleles (e.g. TEF, TRIM63). • 2 type diabetes: typical score (the 62nd percentile in this score; the reference genomic score gives p57).

Reading for immortality: The axis of the nutrient sensors is one of the most favourable points in the genome of Sérgio F: FOXO3 longevity homozygous and low predicted IGF-1, without strong weaknesses in autophagy. This fits in the intermittent rapamycin already underway. Care is not to "load" the axis in the opposite direction (secretagogues GH/IGF-1, isolated leucine) and maintain insulin sensitivity.

2. Method, chain and trust

- **Genotypes:** 33 read directly on the chip (GRCh38, strand +); 34 locally imputed (Beagle 5.5, full genome reference panel 30x, only DR2 $\geq$ 0.8); 5 without data ("SNP not found in the data").
- **Strand:** All alleles were aligned to the GRCh38 strand + and verified against alleles and frequencies of the reference panel. 0 alleles of the literature were in the strand coding – and were converted (complement); 5 palindromic SNPs (A/T or C/G) were resolved by the allelic frequency; the non-resolvable palindromic SNPs were out of the scores.
- **Classification:** compares the copies of the effect allele with the average of the reference population ($2 \times$ frequency). Difference $\geq$ 0.5 copies $\rightarrow$ Favourable or Unfavourable; lower $\rightarrow$ Typical. Not having a rare variant of risk, or a common variant of strong effect, counts as Favourable. An allele with opposite effects (e.g. longer telomeres but higher risk of cancer) appears as Commitment in both lists.
- **Trust:** chip > imputation (DR2 $\geq$ 0.8; estimated accuracy of ~95–99% in common variants, lower in rare ones). Evidence: Strong (replicated in large/functional GWAS), Moderate, Weak (small or inconsistent studies).
- **Polygenic scores:** weighted sum of alleles (published coefficients); percentile calculated against the reference population with exactly the same variants.
- **Sources of interpretation:** dbSNP/Ensembl, ClinVar, GWAS Catalog, PGS Catalog and articles cited in the text.

3. Polygenic scores

Score	Percentile	Variants used	Interpretation	Note
Circulating IGF-1 (Top 52 loci, Sinnott-Armstrong 2021) G_IGF1 · calculated in this report	7	52 of 55 (94%)	Favourable	Low percentile = IGF-1 genetically lower, the profile associated with more longevity in models and in long-lived families (with the possible cost of less muscle anabolism).
Circulating IGF-1 (genomic, 20 829 variants) PGS000687 · calculated in this report	34	20 829 of 23 442 (89%)	Typical	Typical (the 34th percentile) in the same direction as the top loci score.
2 type diabetes (582 variants) PGS000805 · previous genomic imputation (same method)	62	564 of 582 (97%)	Typical	Typical.

the 50th percentile : median of the reference population. For risk scores, high percentile = higher risk; for "pro" scores (e.g. longer telomeres, longevity), high percentile = more of the trait. Typical = percentile between 30 and 70. The scores explain only a small part of the variation and are not diagnostic.

4. Favourable

Gene/score	SNP	Genotype	Source · evidence	Effect
FOXO3	rs2802292 Intron 2 (FOXO3)	GG 2 × G	chip Strong	2 copies of the protective allele G. Allele of more replicated longevity (in several independent cohorts); it is associated with more expression of FOXO3, which activates autophagy, antioxidants and repair. Mechanisms: Deregulation of Nutritional Sensing; Autophagy / Deregulated Macroautophagy; Oxidative stress
FOXO3	rs2764264 intron (FOXO3)	CC 2 × C	chip Moderate	2 copies of the protective allele C. Allele associated with longevity (Willcox 2008 and replicas). Mechanisms: Deregulation of Nutritional Sensing; Autophagy / Deregulated Macroautophagy
FOXO3	rs13217795 intron (FOXO3)	CC 2 × C	chip Moderate	2 copies of the protective allele C. Allele associated with longevity, in linkage imbalance with rs2802292. Mechanisms: Deregulation of Nutritional Sensing; Autophagy / Deregulated Macroautophagy
polygenic	G_IGF1 Circulating IGF-1 (Top 52 loci, Sinnott-Armstrong 2021)	the 7th percentile	52 variants Score	Low percentile = IGF-1 genetically lower, the profile associated with more longevity in models and in long-lived families (with the possible cost of less muscle anabolism). Mechanisms: Deregulation of Nutritional Sensing; Endocrine dysregulation
LINC-PINT, HNF1A-AS1, HMG1P19, PAPPA2, S1PR2, APOH	IGF-1 circulating 14 of 60 loci		GWAS GWAS	14 loci in which the genotype of Sérgio F deviates from the average of the reference population in the favourable sense (favourable = lower IGF-1). Mechanisms: Deregulation of Nutritional Sensing; Endocrine dysregulation

5. Unfavourable

Gene/score	SNP	Genotype	Source · evidence	Effect
CBR3-AS1, TEF, TRIM63, PRSS36, BTBD9	IGF-1 circulating 5 of 60 loci		GWAS GWAS	5 loci in which the genotype of Sérgio F deviates from the mean of the reference population in the unfavourable sense (favourable = lower IGF-1). Mechanisms: Deregulation of Nutritional Sensing; Endocrine dysregulation

6. Rare variants (ClinVar) and chip boundaries

All 5 904 variants classified as pathogenic or probably pathogenic in the ClinVar were searched for in the 42 genes of the panel of this report (AKT1, AKT2, ATG16L1, ATG5, ATG7, BECN1, DEPDC5, EPG5, FOXO1, FOXO3, GH1, GHR, IGF1, IGF1R, IGFALS, INSR, IRGM, IRS1, IRS2, KPTN, LAMTOR2, MTOR, NPRL2, NPRL3, PIK3CA, PRKAA1, PRKAG2, PTEN, RICTOR, RPTOR, SIRT1, SIRT6, SQSTM1, STK11, SZT2, TECPR2, TFEB, TSC1, TSC2, ULK1, WDR45). Of these, 857 have probes on the chip and additional 4 are in the imputation panel.

Gene	Variant	Genotype	Condition (ClinVar)	Evaluation
TSC2	rs45517205 16:2071508 A>C	CC	not provided	Homozygosity for a pathogenic variant of TSC2 would be severe tuberous sclerosis. Artifact.

Chip limit: the chips genotype well common variants, but often fail in the rare ones (false positives and false negatives; Weedon 2021). The absence of pathogenic variants here does not exclude hereditary syndromes. Only clinical sequencing (panel, exome or genome) in a certified laboratory does so.

7. GWAS loci

For each study: independent loci (one SNP per region of 500 kb, the most significant, $p \leq 5 \times 10^{-8}$), with the genotype of Sérgio F and the classification against the reference population. The typical loci are in the complete table (CSV); here we show those that move away from the average.

IGF-1 circulating

Sinnott-Armstrong N et al. 2021 (UK Biobank). Effect allele: Higher IGF-1 = unfavourable. 60 loci: 14 favourable, 5 unfavourable, 35 typical, 6 without data.

Gene (mapped)	SNP	Allele	Genotype	Freq.	Source	Class
LINC-PINT, LINC00513	rs157934	C ↑	TT	0.33	imputation (DR2=1.00)	Favourable
HNF1A-AS1, HNF1A	rs1800574	T ↑	CC	0.03	chip	Favourable
HMGN1P19 - EPS15P1	rs28887535	T ↓	TC	0.12	imputation (DR2=0.99)	Favourable
PAPPA2	rs10913200	A ↑	GG	0.02	chip	Favourable
S1PR2 - LIMASI	rs8105174	T ↓	TC	0.20	imputation (DR2=0.97)	Favourable
CBR3-AS1	rs9636630	G ↓	AA	0.27	imputation (DR2=0.95)	Unfavourable
APOH	rs1801689	C ↑	AA	0.04	chip	Favourable
TMPRSS11E	rs1377870	G ↑	AA	0.29	imputation (DR2=1.00)	Favourable
RPL32P7 - RNU6-1256P	rs9843865	G ↑	GT	0.86	imputation (DR2=0.95)	Favourable
IGFBP3	rs9282734	G ↑	TT	0.00	chip	Favourable
TEF	rs5751086	T ↓	CC	0.27	chip	Unfavourable
TRIM63 - PDIK1L	rs3008233	T ↑	TC	0.19	imputation (DR2=0.99)	Unfavourable
RBFOX1	rs1369924	C ↓	CA	0.18	imputation (DR2=0.99)	Favourable
PRELID1P1 - RPS4XP9	rs74580701	G ↑	AA	0.03	chip	Favourable
PRSS36 - NDUFA3P6	rs12102776	C ↑	CC	0.64	imputation (DR2=0.89)	Unfavourable
ACY1, ABHD14A-ACY1	rs121912698	T ↑	CC	0.00	chip	Favourable
DGKB	rs35221866	T ↓	TC	0.18	imputation (DR2=0.98)	Favourable
TBX3	rs1061657	C ↑	TT	0.28	chip	Favourable
BTBD9	rs2235711	T ↑	TT	0.70	imputation (DR2=0.99)	Unfavourable

8. Map for 84 aging mechanisms

Mechanism	Favourable	Unfavourable	Interpretation
Nutrient Sensing Deregulation	FOXO3 rs2802292; FOXO3 rs2764264; FOXO3 rs13217795; G_IGF1; IGF-1 circulating	IGF-1 circulating	Strong favourable (FOXO3 × 3; Low IGF-1).
Autophagy / Deregulated Macroautophagy	FOXO3 rs2802292; FOXO3 rs2764264; FOXO3 rs13217795	—	FOXO3 favourable; ATG16L1 and IRGM on average.
Oxidative stress	FOXO3 rs2802292	—	Predominant
Endocrine dysregulation	G_IGF1; IGF-1 circulating	IGF-1 circulating	Low predicted IGF-1.
Insulin resistance	—	—	Average heterozygosity (TCF7L2, IRS1); typical 2 type diabetes.

9. Practical section — suggestion (not applied)

Coherent with the existing genetic plan and the current stack reported. Nothing was applied here; everything is to discuss with the doctor.

#	Suggestion	Why (genes)	State
1	Keep rapamycin 4 mg of 6 in 6 weeks (already in the stack)	Intermittent inhibition of mTORC1; compatible with favourable FOXO3. CYP3A5 expresser: a sirolimus dosage helps to calibrate (plan).	suggestion (not applied)
2	10-12 h feed window; prolonged fasting only with the doctor	Activates AMPK/autophagy; with dapagliflozin, long fasts increase the risk of euglycemic ketoacidosis.	suggestion (not applied)
3	Proper protein, distributed, without leucine/BCAA isolates	Do not continuously stimulate mTORC1; the plan already advises BCAA alone.	suggestion (not applied)
4	Resistance and aerobic exercise	Activates AMPK and muscle autophagy and keeps muscle mass with genetically lower IGF-1.	suggestion (not applied)
5	More fiber (more is always better)	Improves insulin sensitivity and intestinal axis (ATG16L1/IRGM on average).	suggestion (not applied)
6	Maintain lithium 5 mg (already in the stack)	Lithium induces autophagy (via inositol and GSK3).	suggestion (not applied)
7	Foods rich in spermidine (wheat germ, mushrooms, legumes)	Natural autophagy inducer, by food.	suggestion (not applied)

Avoiding or with caution

- Secretagogues of GH (GHRP-2/6, CJC-1295, MK-677) and exogenous IGF-1 without medical indication: contradict the favourable profile of low IGF-1. (*suggestion, not applied*)
- Berberin (initiates the CYP3A4 of rapamycin and is added to drugs that lower glucose). (*suggestion, not applied*)

10. Limitations

- The real IGF-1 depends on the diet (protein), age and drugs; the score only predicts the genetic component.
- MTOR and SIRT1 SNPs have weak evidence and above all from other populations.
- The GWAS loci show average associations in the population; the "mapped" gene is the closest and not always the causal gene.
- The effects of isolated SNPs are small; several studies with only one population. They do not allow clinical conclusions by themselves.
- Imputation: an imputed genotype may be wrong, especially in rare variants; the imputed ones are marked.
- This report uses only genes: analysis, exams or measurements were not used.

11. Full table SNP to SNP (cured SNPs)

Gene	SNP	Effect allele (+)	Genotype	Freq.	Source	Chain	Class	Evidence
FOXO3	rs2802292 Intron 2 (FOXO3)	G (protector)	GG	0.434	chip	strand + (without conversion)	Favourable	Strong
FOXO3	rs2764264 intron (FOXO3)	C (protector)	CC	0.370	chip	strand + (without conversion)	Favourable	Moderate
FOXO3	rs13217795 intron (FOXO3)	C (protector)	CC	0.362	chip	strand + (without conversion)	Favourable	Moderate
IGF1R	rs2229765 Glu1043Glu (IGF1R)	A (protector)	AG	0.432	chip	strand + (without conversion)	Typical	Weak
IRS1	rs2943641 near IRS1	C (risk)	CT	0.619	chip	strand + (without conversion)	Typical	Moderate
PPARG	rs1801282 Pro12Ala	G (protector)	CC	0.118	chip	palindromic: confirmed by frequency (reference population 0.12)	Typical	Moderate
FTO	rs9939609 Intron 1 (FTO)	A (risk)	AT	0.420	chip	palindromic: confirmed by frequency (reference population 0.42)	Typical	Strong
TCF7L2	rs7903146 intron (TCF7L2)	T (risk)	TC	0.318	chip	strand + (without conversion)	Typical	Strong
MTOR	rs2295080 promoter (MTOR)	G (protector)	GT	0.314	chip	strand + (without conversion)	Typical	Weak
SIRT1	rs7895833 near SIRT1	G (protector)	AA	0.199	chip	strand + (without conversion)	Typical	Weak
ATG16L1	rs2241880 Thr300Ala	G (risk)	GA	0.549	chip	strand + (without conversion)	Typical	Moderate
IRGM	rs13361189 near IRGM	C (risk)	TT	0.101	chip	strand + (without conversion)	Typical	Moderate

Notes about the typical, informative and without data: IGF1R rs2229765 (AG): The A allele is associated with lower free IGF-1 and was more frequent in long-lived (Bonafé 2003); modest evidence. IRS1 rs2943641 (CT): The C allele is associated with more insulin resistance, less IRS1 and more 2 type diabetes (GWAS). PPARG rs1801282 (CC): The Ala12 allele is associated with higher insulin sensitivity and less 2 type diabetes. FTO rs9939609 (AT): The A allele is associated with more appetite and adiposity; FTO participates in the detection of amino acids by mTORC1. TCF7L2 rs7903146 (TC): The T allele is the largest common genetic sign of 2 type diabetes (less insulin and incretin secretion). MTOR rs2295080 (GT): The G allele reduces the activity of the MTOR promoter and is associated with less risk of some cancers (observational studies). SIRT1 rs7895833 (AA): The G allele was associated with lower adiposity and higher SIRT1 activity in a population study (Zillikens 2009). ATG16L1 rs2241880 (GA): The Ala300 allele makes ATG16L1 more sensitive to cleavage by caspase-3: less autophagy (mainly xenophagy of bacteria) and more risk of Crohn's disease. IRGM rs13361189 (TT): The C allele marks the deletion that alters the expression of IRGM (autophagy of intracellular bacteria); it is associated with Crohn's disease.